

A Characterization of Interior APEPs in ℓ_p -Sums

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns Question 8.2 from:

Ramon J. Aliaga, Luis C. Garcia-Lirola, Juan Guerrero-Viu, Matias Raja, Abraham Rueda Zoca, *Almost preserved extreme points*, arXiv:2510.21234.

The paper asks for a characterization of interior almost preserved extreme points in ℓ_p -sums. The source location is page 37 of the paper.

For the case of ℓ_p -sums, in Proposition 4.9 we have characterised when the norm-one elements are APEP of the unit ball. However, we were unable to describe those APEP whose norm is strictly smaller than 1.

Question 8.2. Let X be the ℓ_p -sum of a family $\{X_i : i \in I\}$ of Banach spaces, where $1 < p < \infty$, and let $(x_i) \in B_X$ with $\|(x_i)\| < 1$. When is (x_i) an APEP of B_X ?

Even though we do not have a complete characterisation, let us now present a sufficient condition concerning the above question.

We use two results from the source paper. First, if C is a closed, bounded, convex subset of a Banach space X , then

$$x \in \text{APEP}(C) \iff x \in \overline{\text{ext } C}^{w^*} \subset X^{**}.$$

Second, for $1 < p < \infty$,

$$\text{ext } B_{(\oplus_i E_i)_p} = \left\{ (u_i) \in S_{(\oplus_i E_i)_p} : u_i = 0 \text{ or } u_i / \|u_i\| \in \text{ext } B_{E_i} \text{ for every } i \right\}.$$

Notation

Let $1 < p < \infty$, let $\{X_i : i \in I\}$ be a family of Banach spaces, and let

$$X = \left(\bigoplus_{i \in I} X_i \right)_p.$$

For each i , put

$$C_i = \overline{\text{ext } B_{X_i^{**}}}^{w^*} \subset B_{X_i^{**}}.$$

For $u \in X_i$, define the set of *admissible radii*

$$R_i(u) = \{r \in [0, 1] : u \in rC_i\},$$

where $0C_i = \{0\}$. If $R_i(u) \neq \emptyset$, define

$$\rho_i(u) = \min R_i(u).$$

If $R_i(u) = \emptyset$, set $\rho_i(u) = +\infty$. Finally let

$$K = \{i \in I : X_i \neq \{0\}\}.$$

If $K = \emptyset$, then $X = \{0\}$ and the only point of B_X is an APEP. The theorem below treats the nontrivial case $K \neq \emptyset$.

Lemma 1. *For each i and $u \in X_i$, the set $R_i(u)$ is compact. In particular, the minimum defining $\rho_i(u)$ exists whenever $R_i(u)$ is nonempty.*

Proof. The set C_i is weak-star compact. Suppose $r_\alpha \in R_i(u)$ and $r_\alpha \rightarrow r$ in $[0, 1]$. Choose $c_\alpha \in C_i$ with $u = r_\alpha c_\alpha$. Passing to a subnet, $c_\alpha \rightarrow c \in C_i$ in the weak-star topology. Then $u = rc$, so $r \in R_i(u)$. $\square$

Characterization

Theorem 1. *Let $x = (x_i) \in B_X$ with $\|x\| < 1$.*

1. *If K is nonempty and finite, then $x \in \text{APEP}(B_X)$ if and only if there exist radii $r_i \in R_i(x_i)$, $i \in K$, such that*

$$\sum_{i \in K} r_i^p = 1.$$

2. *If K is infinite, then $x \in \text{APEP}(B_X)$ if and only if*

$$\sum_{i \in I} \rho_i(x_i)^p \leq 1.$$

Proof. We identify

$$X^{**} = \left(\bigoplus_{i \in I} X_i^{**} \right)_p \quad \text{and} \quad X^* = \left(\bigoplus_{i \in I} X_i^* \right)_q,$$

where q is conjugate to p .

Necessity. Assume $x \in \text{APEP}(B_X)$. By the quoted characterization, there is a net $y^\alpha = (y_i^\alpha) \in \text{ext } B_{X^{**}}$ converging weak-star to x . Fix a finite set $F \subset I$. By compactness of $[0, 1]^F$, pass to a subnet so that

$$\|y_i^\alpha\| \rightarrow r_i \quad (i \in F).$$

If $r_i = 0$, then $x_i = 0$, so $r_i \in R_i(x_i)$. If $r_i > 0$, then eventually $y_i^\alpha \neq 0$, and

$$e_i^\alpha = \frac{y_i^\alpha}{\|y_i^\alpha\|} \in \text{ext } B_{X_i^{**}}.$$

Passing to a further subnet, $e_i^\alpha \rightarrow e_i \in C_i$ weak-star, and $x_i = r_i e_i$. Thus $r_i \in R_i(x_i)$ and $\rho_i(x_i) \leq r_i$. Since each y^α has norm one,

$$\sum_{i \in F} \rho_i(x_i)^p \leq \sum_{i \in F} r_i^p = \lim_\alpha \sum_{i \in F} \|y_i^\alpha\|^p \leq 1.$$

Taking the supremum over finite F gives

$$\sum_{i \in I} \rho_i(x_i)^p \leq 1.$$

This proves the necessity in the infinite case.

If K is finite, apply the same argument with $F = K$. Since all nonzero summands are indexed by K , the limiting radii satisfy

$$\sum_{i \in K} r_i^p = \lim_\alpha \sum_{i \in K} \|y_i^\alpha\|^p = 1,$$

and each r_i belongs to $R_i(x_i)$.

Sufficiency when K is finite. Assume there are $r_i \in R_i(x_i)$ with $\sum_{i \in K} r_i^p = 1$. Let

$$U = \{z \in B_{X^{**}} : |f^k(z - x)| < \varepsilon, \ k = 1, \dots, m\}$$

be a basic weak-star neighborhood of x . For every $i \in K$ with $r_i > 0$, choose $c_i \in C_i$ with $x_i = r_i c_i$. Since c_i belongs to the weak-star closure of $\text{ext } B_{X_i^{**}}$, we may choose $e_i \in \text{ext } B_{X_i^{**}}$ so close to c_i on the finitely many coordinate functionals $f_i^1, \dots, f_i^m$ that the vector y defined by

$$y_i = \begin{cases} r_i e_i, & r_i > 0, \\ 0, & r_i = 0 \end{cases}$$

satisfies $|f^k(y - x)| < \varepsilon$ for all k . The equality $\sum_{i \in K} r_i^p = 1$ and the extreme-point formula show that $y \in \text{ext } B_{X^{**}}$. Hence every weak-star neighborhood of x meets $\text{ext } B_{X^{**}}$, so $x \in \text{APEP}(B_X)$.

Sufficiency when K is infinite. Assume $\sum_i \rho_i(x_i)^p \leq 1$. Let U be a basic weak-star neighborhood of x as above. Because $q < \infty$, finite-support functionals are norm dense in $X^* = (\bigoplus_i X_i^*)_q$. Choose finite-support $g^1, \dots, g^m \in X^*$, supported on a common finite set F , such that

$$\|f^k - g^k\| < \varepsilon/4 \quad (k = 1, \dots, m).$$

Put

$$s = \sum_{i \in F} \rho_i(x_i)^p.$$

Then $s \leq 1$.

For each $i \in F$ with $\rho_i(x_i) > 0$, choose $c_i \in C_i$ such that $x_i = \rho_i(x_i) c_i$, and then choose $e_i \in \text{ext } B_{X_i^{**}}$ close enough to c_i on the finitely many coordinate functionals $g_i^1, \dots, g_i^m$ that

$$\sum_{i \in F} |g_i^k(\rho_i(x_i) e_i - x_i)| < \varepsilon/2 \quad (k = 1, \dots, m).$$

If $\rho_i(x_i) = 0$, set the corresponding coordinate equal to zero.

If $s < 1$, choose $i_0 \in K \setminus F$ and choose an extreme point $e_{i_0} \in \text{ext } B_{X_{i_0}^{**}}$; this is possible by Krein-Milman because $X_{i_0} \neq \{0\}$. Put $r_0 = (1 - s)^{1/p}$. Define $y \in X^{**}$ by

$$y_i = \begin{cases} \rho_i(x_i)e_i, & i \in F, \rho_i(x_i) > 0, \\ r_0 e_{i_0}, & i = i_0, s < 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $s = 1$, use the same definition but omit the i_0 line. In both cases $\|y\| = 1$, and every nonzero normalized coordinate is an extreme point of the corresponding coordinate ball. Hence $y \in \text{ext } B_{X^{**}}$.

For each k ,

$$|g^k(y - x)| = \left| \sum_{i \in F} g_i^k(y_i - x_i) \right| < \varepsilon/2.$$

Also $\|y - x\| \leq 2$, so

$$|(f^k - g^k)(y - x)| \leq \|f^k - g^k\| \|y - x\| < \varepsilon/2.$$

Thus $|f^k(y - x)| < \varepsilon$ for every k , and $y \in U \cap \text{ext } B_{X^{**}}$. The weak-star closure characterization therefore gives $x \in \text{APEP}(B_X)$. $\square$

Checks and Consequences

If $\|x\| = 1$, then $\sum_i \|x_i\|^p = 1$ and $\rho_i(x_i) \geq \|x_i\|$. The theorem therefore forces $\rho_i(x_i) = \|x_i\|$ for all nonzero coordinates, which is exactly the source paper's norm-one characterization: each nonzero normalized coordinate must be an APEP in its coordinate ball.

The earlier sufficient condition in Proposition 8.3 of the source paper is recovered by taking the original coordinate radii on the nonzero support and using an admissible positive radius at a zero coordinate where 0 is APEP. The infinite-summand partial result from the previous packet is the special case where $\rho_i(x_i) = \|x_i\|$ for every nonzero coordinate and the leftover mass is put outside the finite weak-star test.

Verification Notes

No computational verifier was used. The argument is structural and rests on weak-star compactness, finite-support approximation in $(\bigoplus_i X_i^*)_q$, and the extreme-point formula for ℓ_p -sums.

Human review should focus on the distinction between the finite and infinite nonzero-summand cases: finite sums require admissible radii whose p -sum is exactly one, while infinite sums require only that the least admissible radii have p -sum at most one.

Limitations and Novelty Check

This is a candidate full answer to Question 8.2 as stated. It is an abstract characterization rather than a closed-form list for every coordinate space; the coordinate information is encoded in the weak-star closures $C_i = \overline{\text{ext } B_{X_i^{**}}}^{w^*}$, which are already central in the source paper's APEP characterization.

Human Review: Interior APEPs in ℓ_p -sums

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed characterization appears correct and quite natural. The source paper gives the general criterion

$$x \in \text{ape}(C) \iff x \in \overline{\text{ext}(C^{w*})}^{w*},$$

and Proposition 4.9 characterizes the norm-one APEPs in ℓ_p -sums. Question 8.2 asks for the remaining interior case. This packet supplies a full characterization by introducing the admissible coordinate radii.

Main Idea

For each coordinate, set

$$C_i = \overline{\text{ext } B_{X_i^{**}}}^{w*}.$$

For $u \in X_i$, a radius $r \in [0, 1]$ is admissible if

$$u \in rC_i.$$

The least admissible radius $\rho_i(u)$ measures how much p -mass is needed in the i -th coordinate to realize u as a weak-star limit of scaled coordinate extreme points.

With this notation, the characterization is very natural. In the infinite summand case, $x = (x_i)$ is an interior APEP precisely when

$$\sum_i \rho_i(x_i)^p \leq 1.$$

The remaining p -mass can be placed in a coordinate outside any finite weak-star test. In the finite summand case, there is no outside coordinate, so one needs admissible radii $r_i \in R_i(x_i)$ with exact equality

$$\sum_i r_i^p = 1.$$

Proof Check

The necessity argument is sound. If x is an APEP, Theorem 3.8 gives a net of extreme points

$$y^\alpha = (y_i^\alpha) \in \text{ext } B_{X^{**}}$$

weak-star converging to x . Coordinatewise weak-star convergence follows from the coordinate embeddings $X_i^* \hookrightarrow X^*$. For each finite set of coordinates, one passes to a subnet so that

$$\|y_i^\alpha\| \rightarrow r_i.$$

If $r_i = 0$, then $x_i = 0$, hence $0 \in R_i(x_i)$. If $r_i > 0$, then

$$\frac{y_i^\alpha}{\|y_i^\alpha\|} \in \text{ext } B_{X_i^{**}},$$

and after passing to a further subnet one obtains $x_i = r_i e_i$ with $e_i \in C_i$. Thus $r_i \in R_i(x_i)$ and $\rho_i(x_i) \leq r_i$. Since every y^α has norm one, finite partial sums satisfy

$$\sum_{i \in F} \rho_i(x_i)^p \leq 1.$$

Taking the supremum over finite F gives the required inequality in the infinite case, while taking F to be the finite support gives the equality condition in the finite case.

The sufficiency argument is also sound. In the finite case, admissible radii with p -sum equal to one allow one to approximate each coordinate by genuine coordinate extreme points and thereby build global extreme points of $B_{X^{**}}$ weak-star close to x . In the infinite case, finite-support approximation in

$$X^* = \left(\bigoplus_i X_i^* \right)_q$$

reduces a weak-star neighbourhood to finitely many coordinates. One realizes the required coordinates using the least admissible radii, and if there is leftover mass, places it in an unused coordinate. The extreme-point formula for ℓ_p -sums then gives an extreme point of $B_{X^{**}}$ inside the chosen neighbourhood.

Review Outcome

Approve this packet as an apparently correct full solution to Question 8.2. The characterization is not simply copied from the paper: the paper supplies the general weak-star closure criterion and the norm-one ℓ_p -sum case, while this argument extends the analysis to interior points through the admissible-radius formulation.